5.2.1 Heat

1. **Tyndallisation:** Rapid steaming at 100°C of culture media n each of three successive days, allowing spores to germinate and to be subsequently killed.
2. Boiling: This method is suitable for enamel metal, glass and rubber wares.

Bowl sterilizers are used for large articles.

Instrument sterilizers are used for smaller articles.

Points to be remember:

- See that articles are quite clean and completely immersed in the water which also must be clean.
- Only after the water comes to the boil start timing. If more articles are added the sterilization to be must begin again
- Boil vigorously for 5 minutes boiling will not kill spores.
- Remove articles with sterile chattel or other lifting forceps on to a sterile surface.

3. Autoclaving: This is a reliable method. This is the method used for most articles.



Points to remember:

- All the articles should be clean and dry before packing.
- The hole in the drums must be open with placing into the autoclave and closed immediately on taking them out.
- Bundle should be not too large and not tightly packed. Steam should be able to penetrate to the centre.
- Rubber gloves should in 5 lbs for 15 mins.
- To autoclave bottles of fluid, loosen the screw caps. Evacuate the steam slowly.

General instructions:

- The articles being sterilized should with stand high temperature.
- The wrapper and the container should allow penetration of the steam into the article.
- The inner champers must not be too full nor the contents arranged too compactly.
- The temperature and pressure of the steam should be high enough to kill all the micro organisms including spores.
- The destruction of a bacteria depends upon the length of time the articles are exposed to steam under pressure. If the time is increased to the exposure the pressure is reduced.
- In operating an autoclave, it is important to remember that all the air in the inner chamber must be driven out and entirely replaced by steam.
- The articles should be left in the autoclave for a short times after the procedure is over to dry materials.

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